

M-CATS

UNIVERSITY OF KENTUCKY

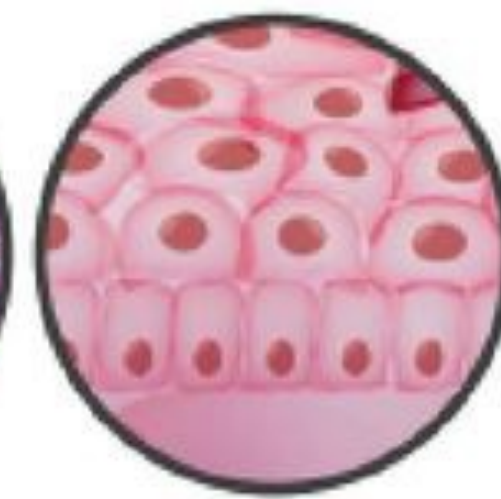
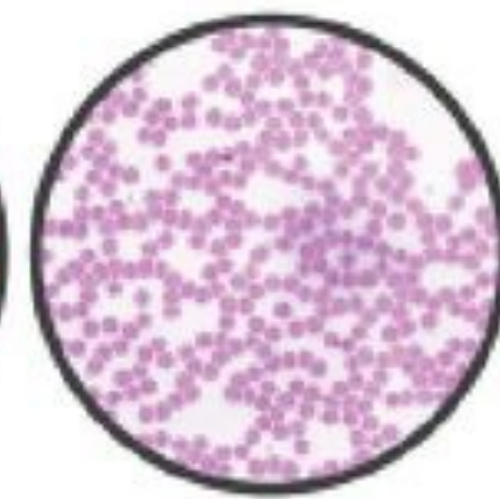
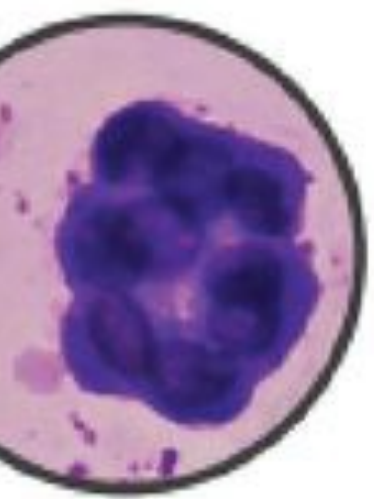
THE M-CATS PRESENTS

Embryogenesis

BIOLOGY CHAPTER 3

ANMOL N, ZANDER H, NIKHKIL K

I (women in) STEM Cells



Lung

Pancreas

Heart Muscle

Red blood cells

Skin

N



Multipotent stem cells



Pluripotent embryonic stem cells

Endoderm line

Mesoderm line

Ectoderm line



Multipotent stem cells

Totipotent embryonic stem cell



Pluripotent embryonic stem cells

Human em
Ind

Endoderm line

Mesoderm line

Ectoderm line

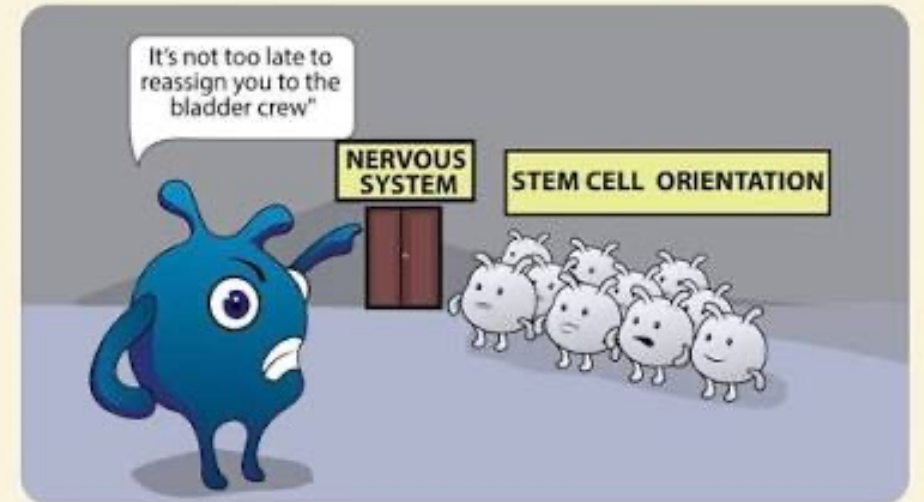
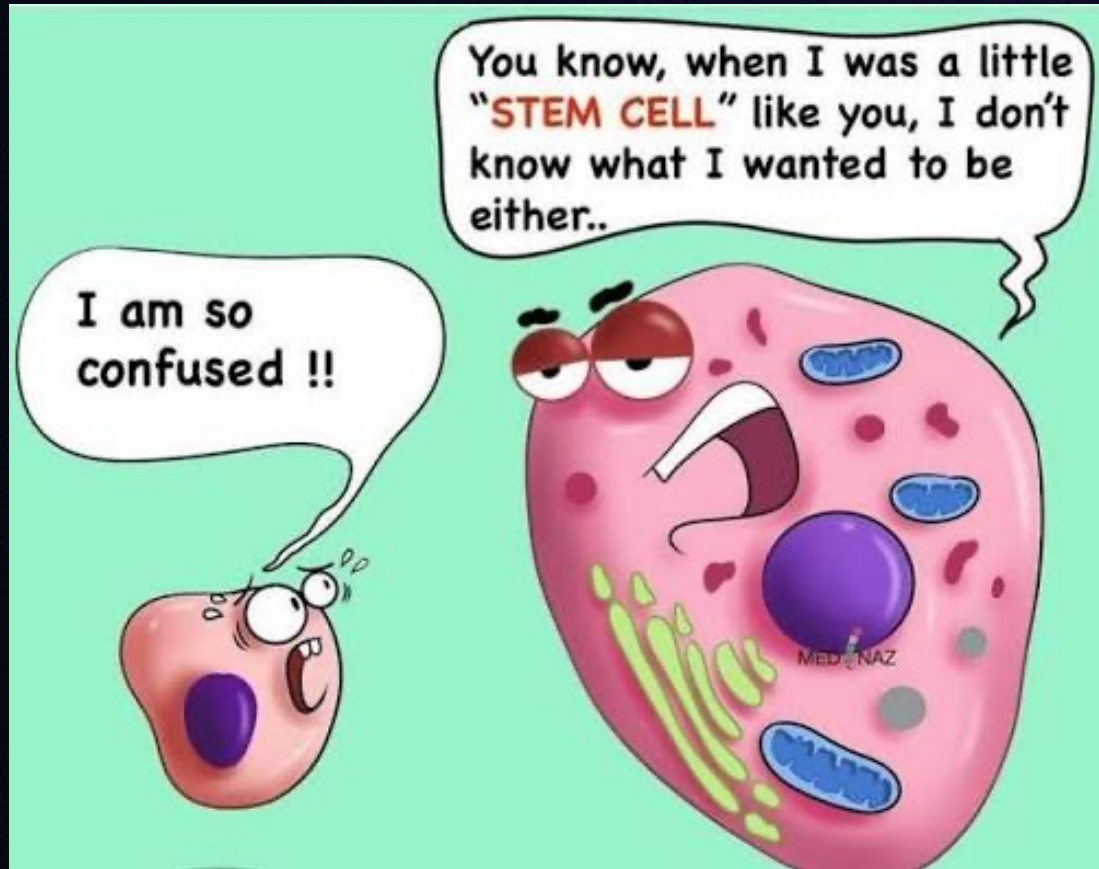
Specification

Determination

Differentiation

Stem Cell Therapy

Inducers



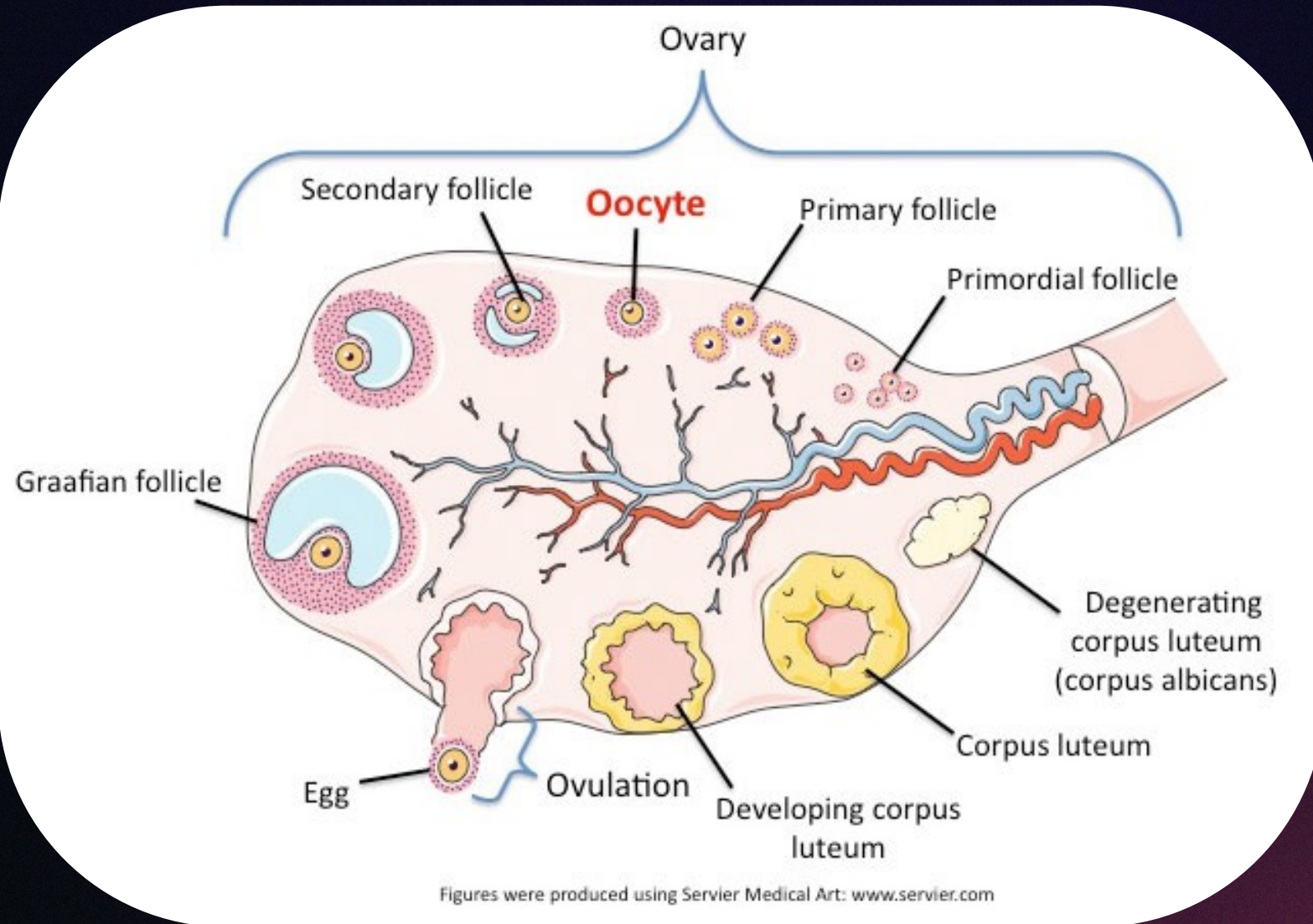
A researcher isolates adult stem cells from a patient's bone marrow. These cells are found to differentiate into red blood cells, white blood cells, and platelets, but cannot differentiate into neurons, hepatocytes, or skin cells. The researcher then treats a separate batch of these same cells with specific transcription factors in an attempt to increase their developmental potential. Which of the following statements accurately describe(s) this scenario?

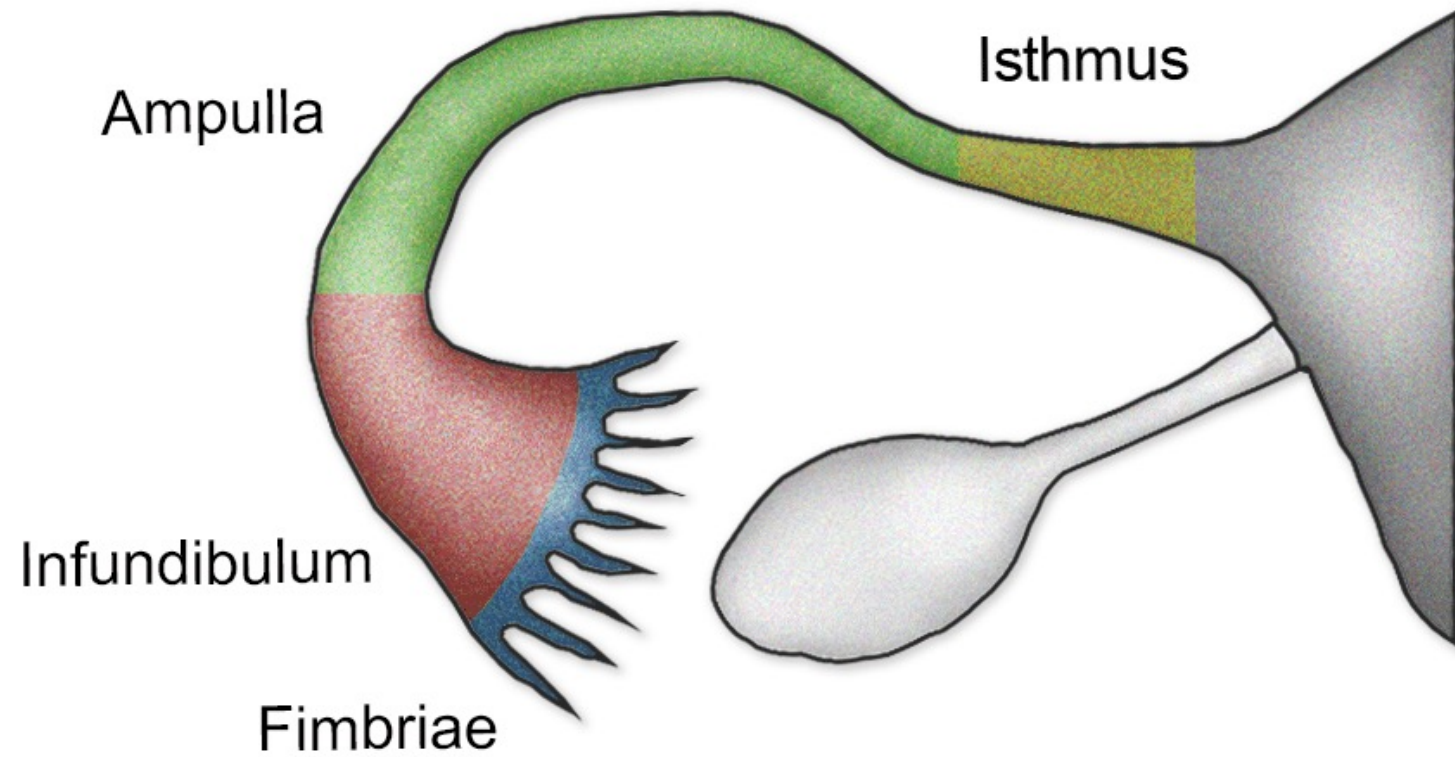
- I. The bone marrow stem cells, as originally isolated, are best classified as multipotent
- II. Using these cells (rather than embryonic stem cells) for a future transplant into the same patient would carry a lower risk of immune rejection
- III. Once these cells have differentiated into blood cell types, they remain capable of reversing this commitment and returning to a totipotent state

- a. I only
- b. I and II
- c. II and III
- d. I, II, and III

- II -

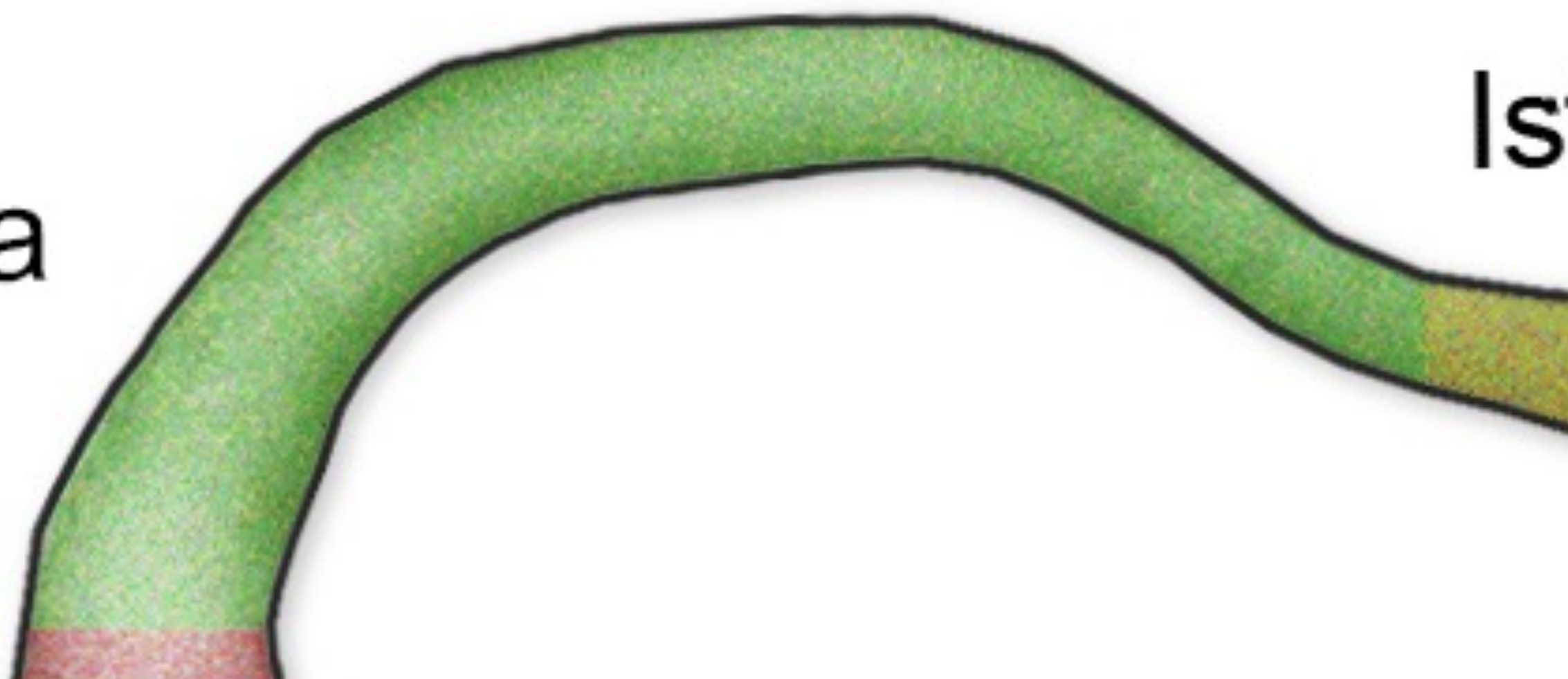
Babyogenesis

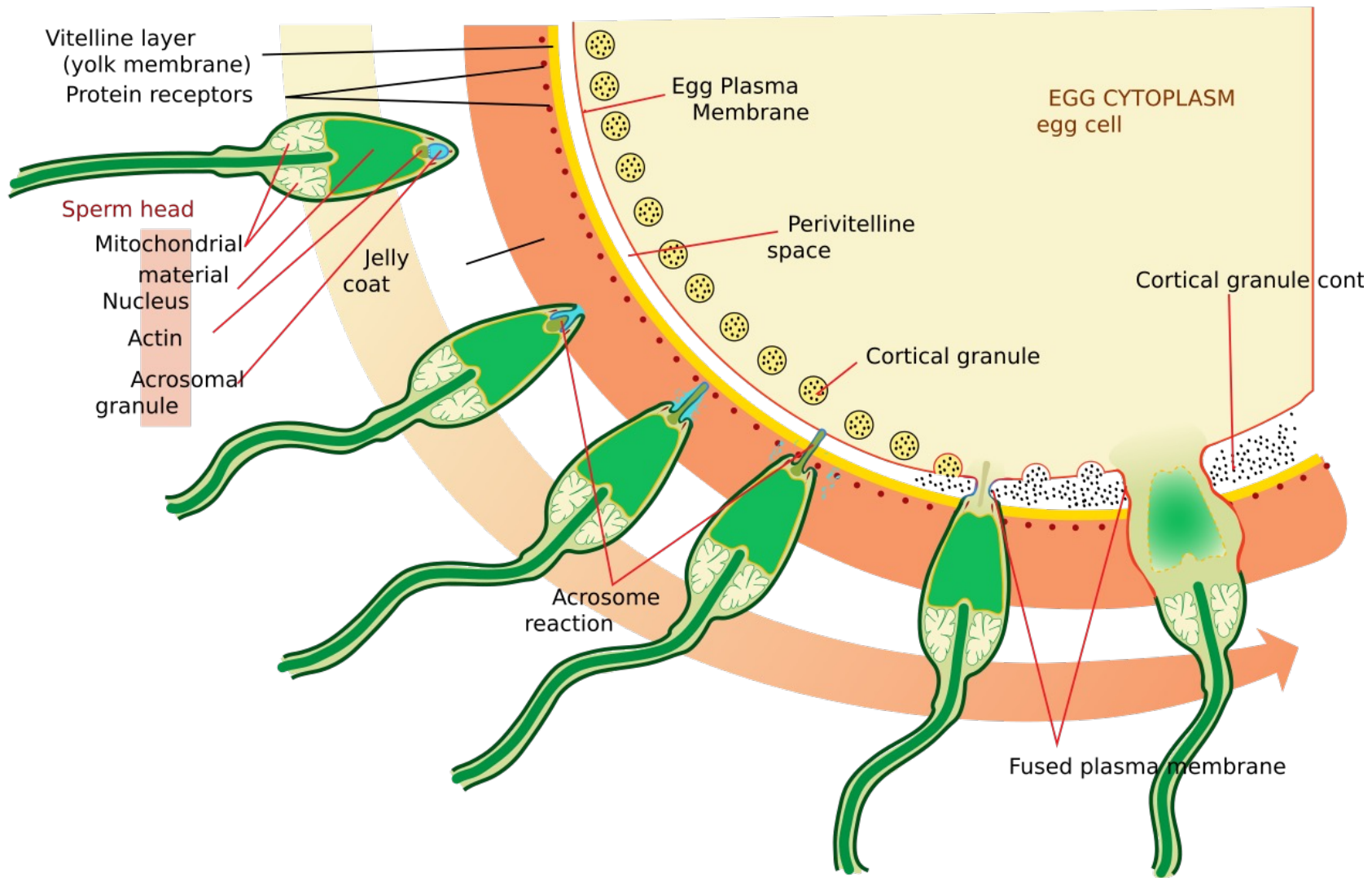


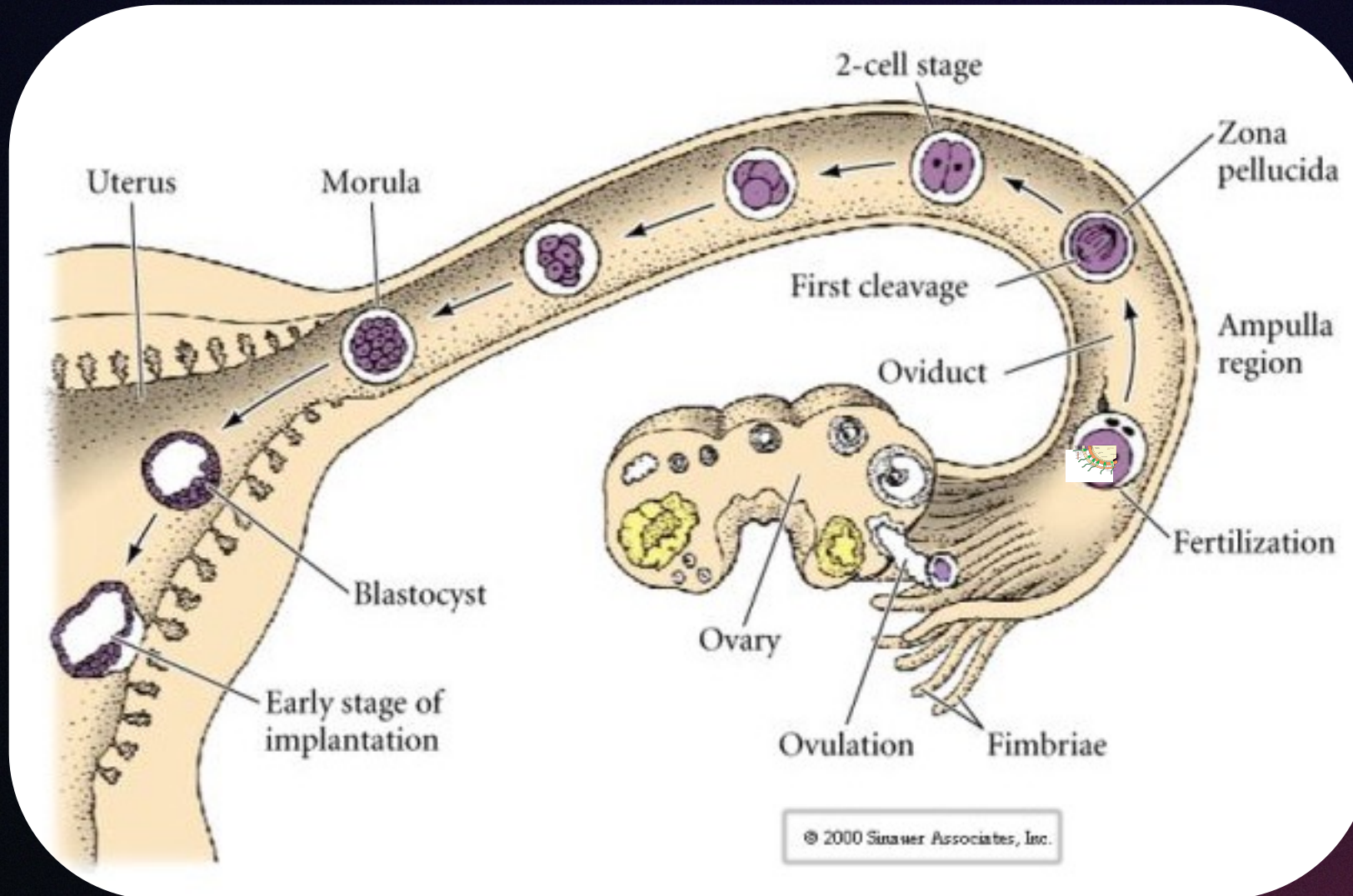


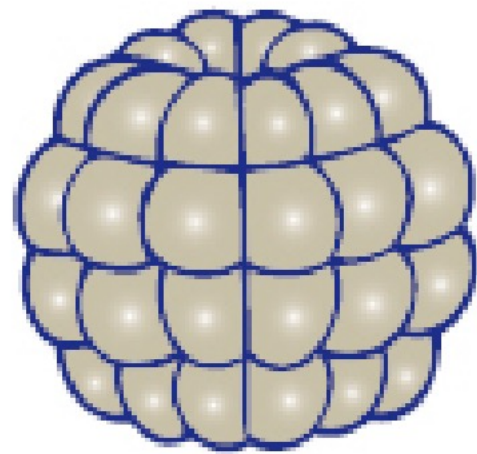
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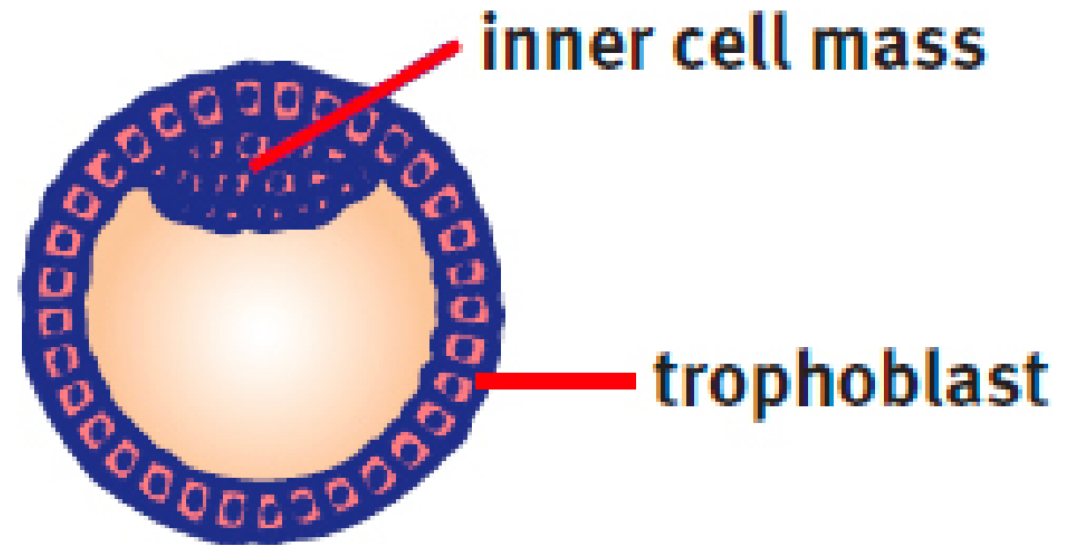


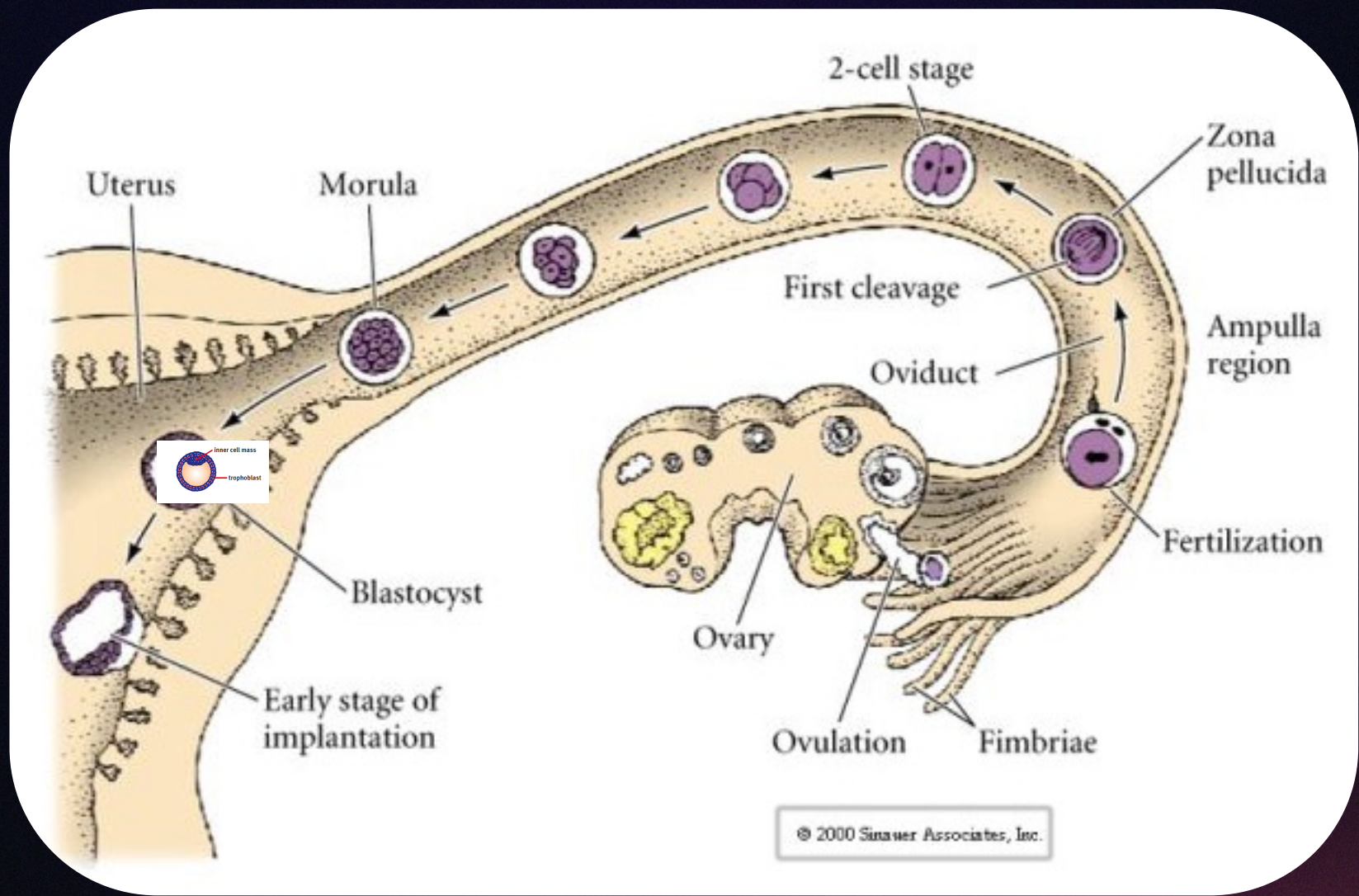






morula





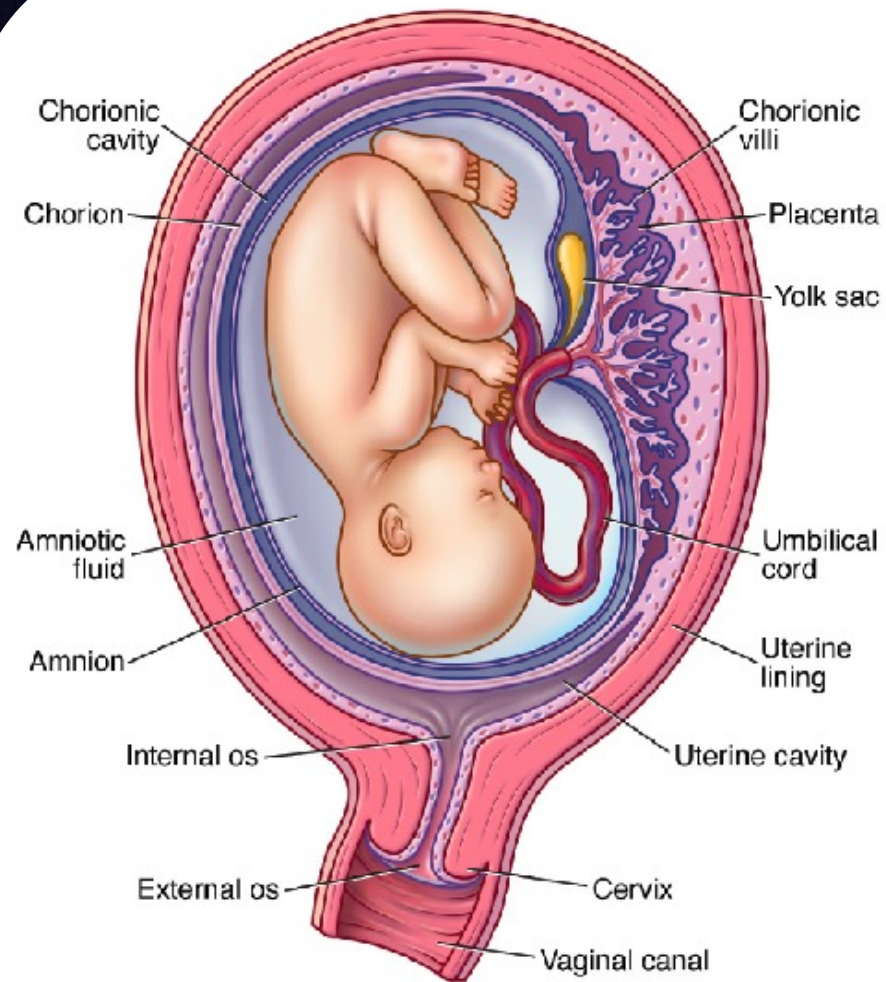
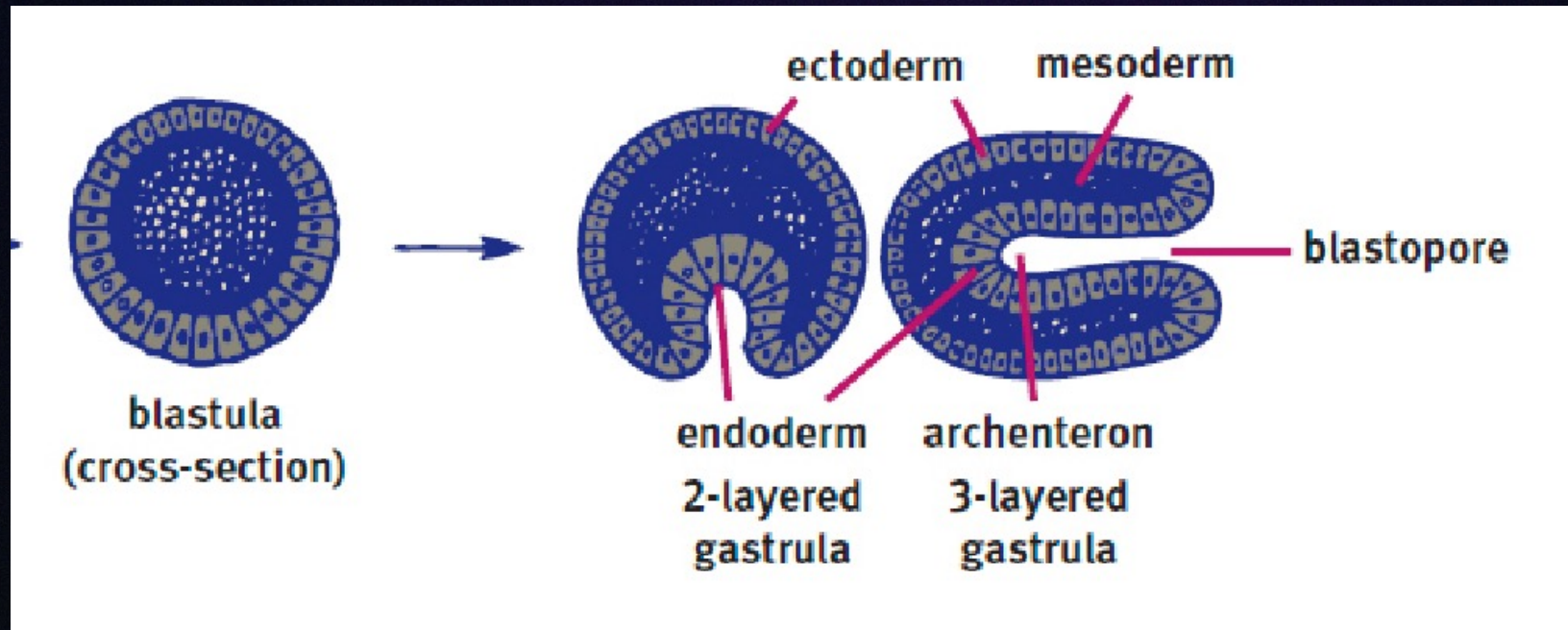


Figure 3.5 Anatomy of Pregnancy

Gastrulation

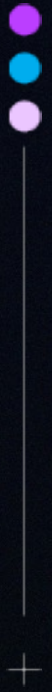


3 layers

Ectoderm

Mesoderm

Endoderm



Derm Layers

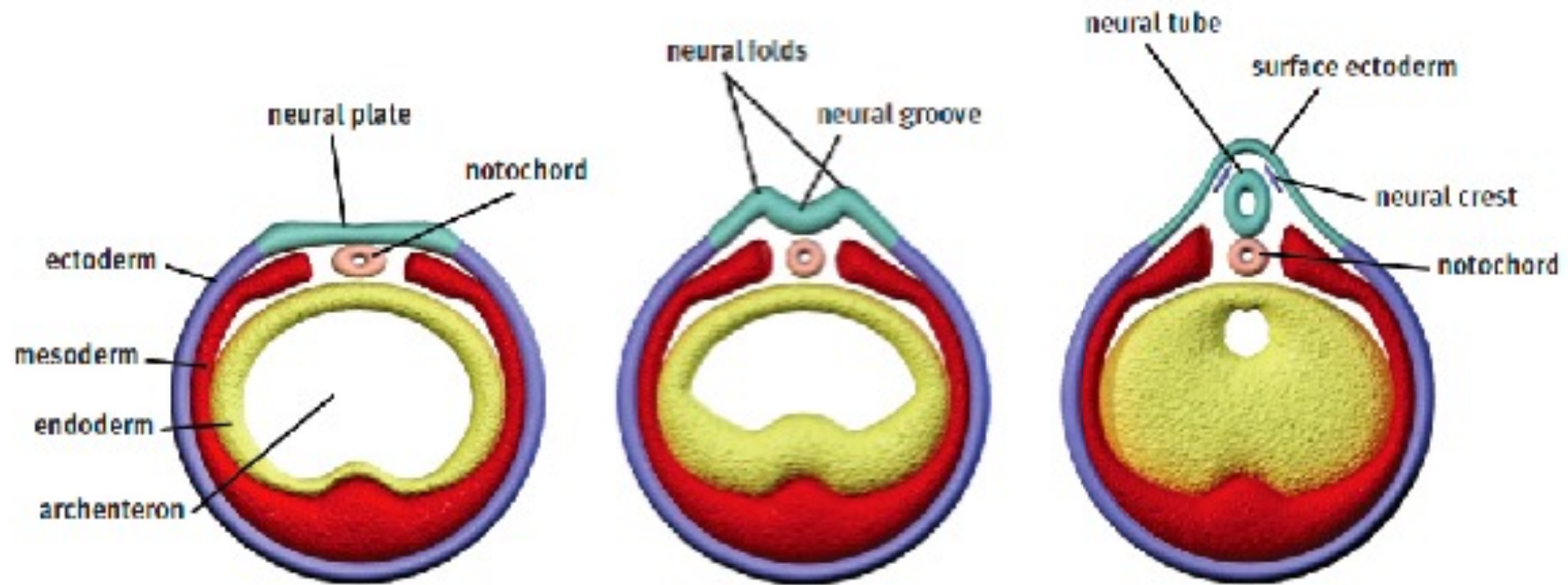
Ectoderm Mesoderm Endoderm

Fill It in!



Differentiation

Neurulation





Teratogens



Hol up'

Teratogens

Wut uhbout twihns doh

Concept Check

A researcher studying sea urchin development observes a blastula undergoing invagination during gastrulation, ultimately forming an archenteron and a blastopore. Which of the following statements about this process is/are accurate?

- I. The blastocoel is progressively displaced/eliminated as the archenteron forms
- II. The archenteron will eventually develop into the gut
- III. If this organism is a deuterostome, the blastopore observed in this experiment will become the mouth

- a) I and II
- b) I and III
- c) II and III
- d) I, II, and III

Concept Check

A researcher observes that a fertilized ovum's plasma membrane undergoes depolarization within seconds of sperm penetration, mediated by a release of calcium ions. What is the primary functional purpose of this event?

- 1.To increase the size of the zygote in preparation for cleavage
- 2.To prevent polyspermy by blocking additional sperm from fertilizing the ovum
- 3.To trigger the completion of meiosis I in the oocyte
- 4.To initiate the acrosomal reaction in the fertilizing sperm

Concept Check

The adrenal gland is composed of two distinct regions: the adrenal cortex and the adrenal medulla. Identify the germ layer origin of each region, and explain why this dual origin makes biological sense given the function of the adrenal medulla.

Concept Check

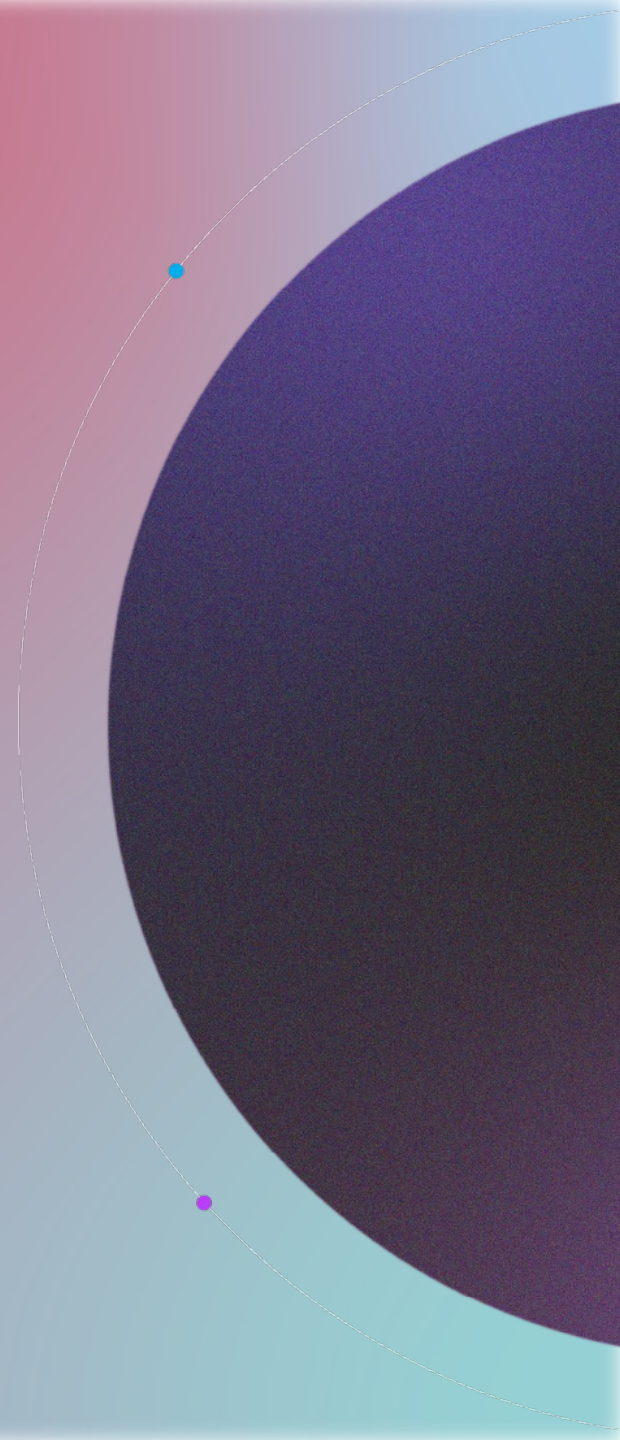
A newborn is found to have a constellation of seemingly unrelated abnormalities: patchy areas of unpigmented skin, dangerously low blood pressure during stress that fails to improve with typical epinephrine release, abnormal calcium regulation traced to malfunctioning thyroid cells, and poor myelination of peripheral nerves. Genetic testing reveals a single mutation affecting a transcription factor required for the proper migration and differentiation of one specific embryonic cell population. Which of the following is most likely the affected cell population?

1. Mesodermal cells,
2. Endodermal cells,
3. Neural Crest cells
4. Trophoblast cells.

- II -

Mechanisms of Development

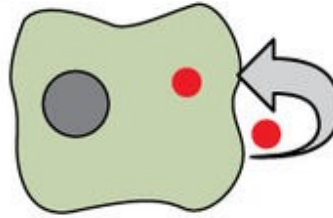
(highkey Kaplan organized this
pretty badly)



Forms of Chemical Signaling

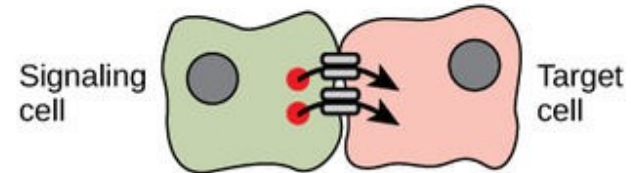
Autocrine

A cell targets itself.



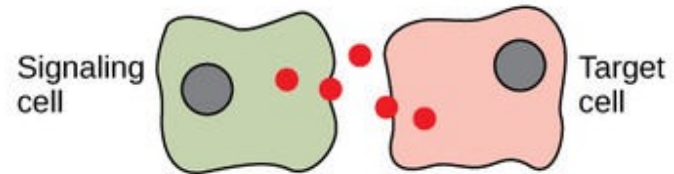
Signaling across gap junctions

A cell targets a cell connected by gap junctions.



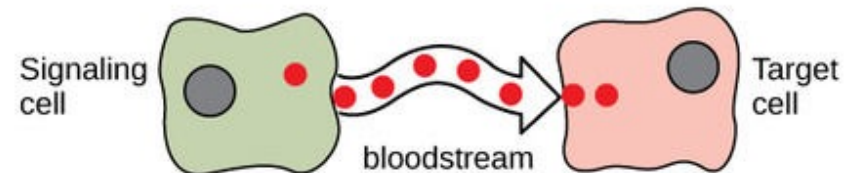
Paracrine

A cell targets a nearby cell.



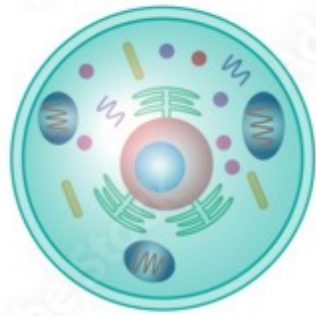
Endocrine

A cell targets a distant cell through the bloodstream.



Apoptosis

- programmed cell death -



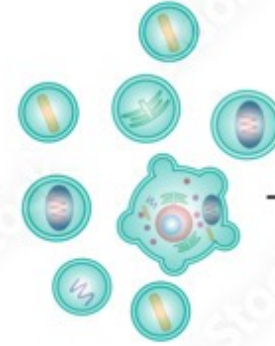
Normal cell



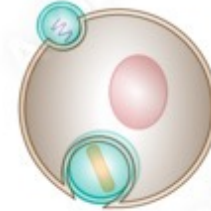
Shrinkage



Membrane
blebbing



Cell breaks into
apoptotic bodies

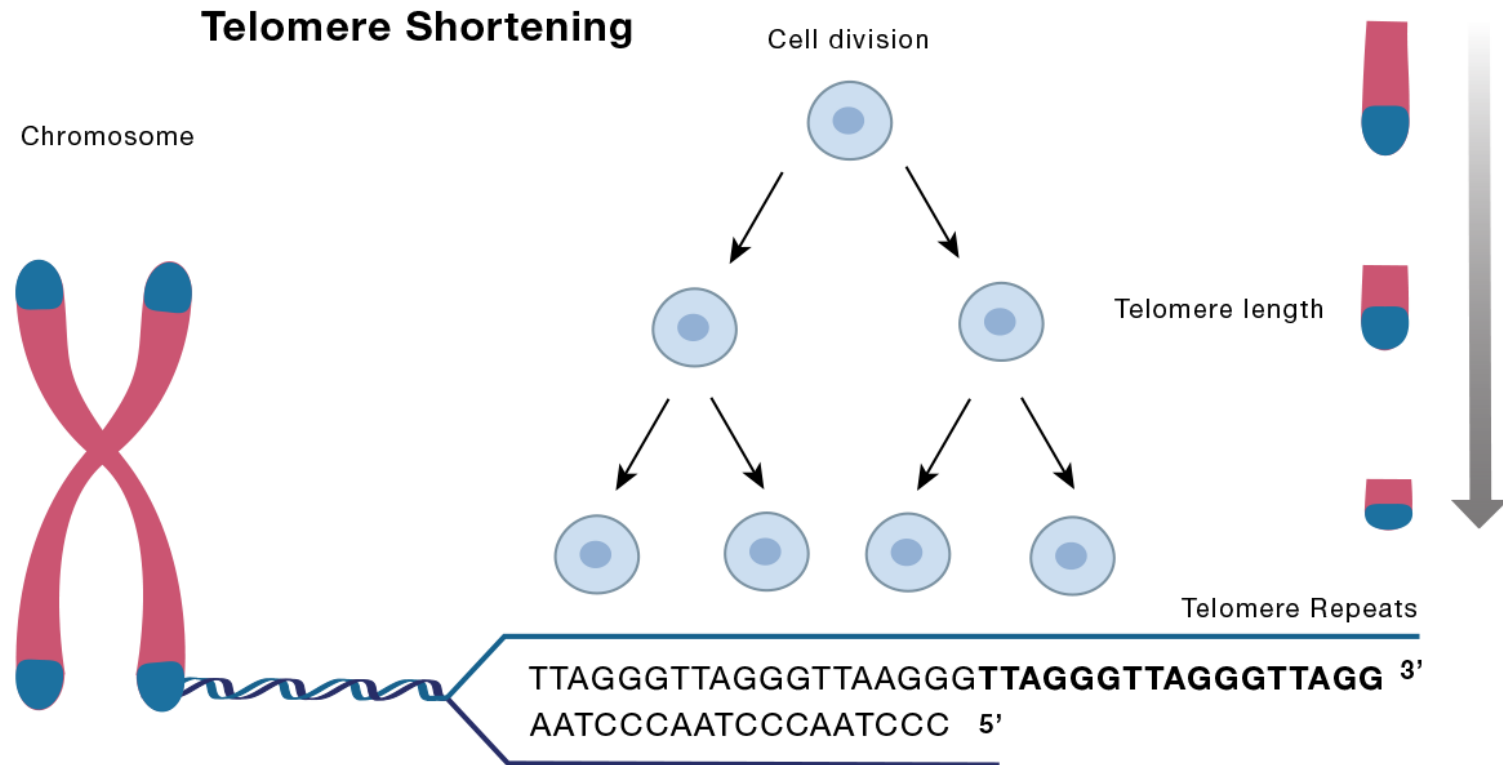


Phagocytosis

vs. Necrosis

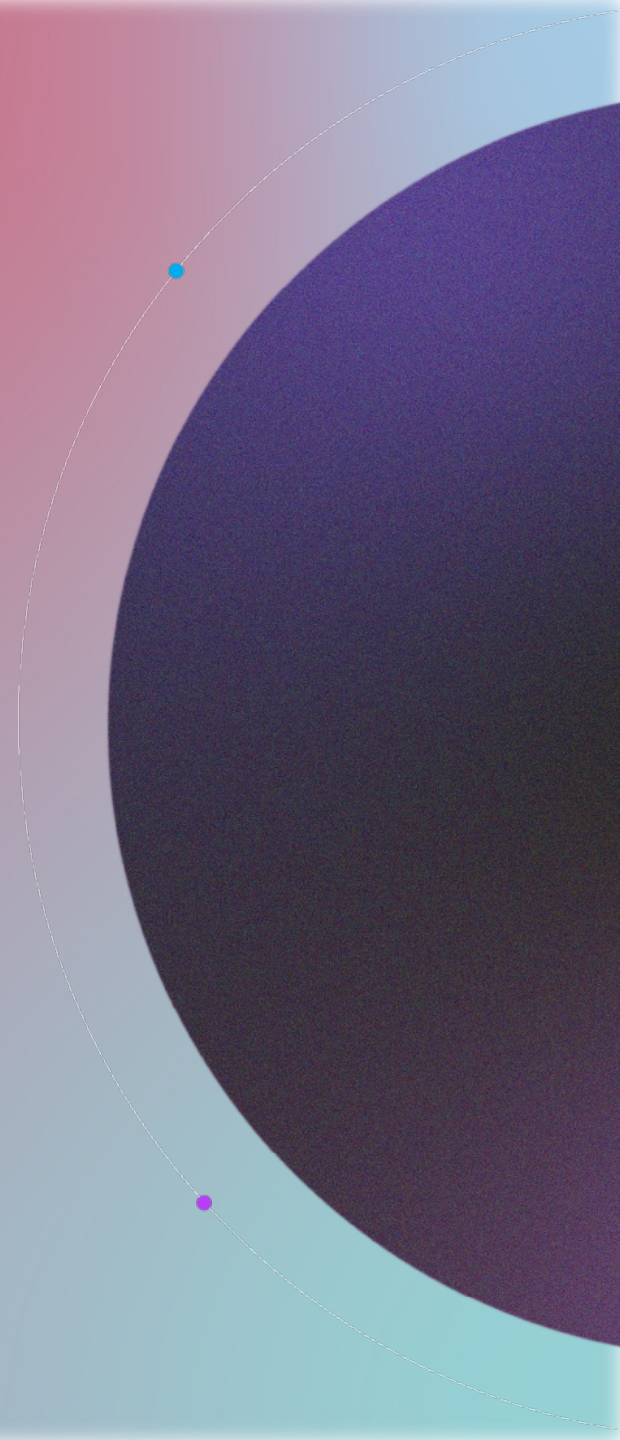
Complete vs Incomplete Regeneration

Telomeres and Senescence



- III -

Fetal Circulation & Preganant



Fetal Circulation

Pregnancy